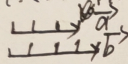
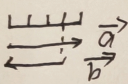


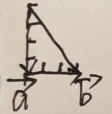
7. (a) They have the same direction:

 $|\vec{a} + \vec{b}| = 7\text{m}$

(b) They have the contrary direction:

 $|\vec{a} + \vec{b}| = 1\text{m}$

(c) They are at right angle:

 $|\vec{a} + \vec{b}| = 5\text{m}$

32. $\tan \theta = \frac{|\vec{b}|}{|\vec{a}|} \Rightarrow \theta \approx 37^\circ$

(a) $|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin 90^\circ = 12$

(b) points out of the paper

(c) $|\vec{a} \times \vec{c}| = |\vec{a}| |\vec{c}| \sin(180^\circ - \theta) = 12$

(d) points ~~into~~ into the paper

(e) $|\vec{b} \times \vec{c}| = |\vec{b}| |\vec{c}| \sin(90^\circ - \theta) = 12$

(f) points out of the paper.

15. $x_a = |\vec{a}| \cos \theta = 5.5\text{m}$, $y_a = |\vec{a}| \sin \theta = 4.5\text{m}$

$x_b = |\vec{b}| \cos(\theta + \theta_2) = -5.5\text{m}$, $y_b = |\vec{b}| \sin(\theta + \theta_2) = 4.5\text{m}$

(a) $x_r = x_a + x_b \approx 0.39\text{m}$ (b) $y_r = y_a + y_b = 12.07\text{m}$

(c) $|\vec{r}| = \sqrt{x_r^2 + y_r^2} \approx 12.07\text{m}$

(d) $\tan \theta = \frac{y_r}{x_r} \approx 37.72^\circ$

(c) $|\vec{r}| = \sqrt{x_r^2 + y_r^2} \approx 12.2\text{m}$

(d) $\tan \theta = \frac{y_r}{x_r} \Rightarrow \theta = 82.5^\circ$

37. (a) $\vec{a} \cdot (\vec{b} \times \vec{c}) = (3.0\hat{i} + 3.0\hat{j} - 2.0\hat{k}) \cdot (-4.0\hat{i} + 1.0\hat{j} + 1.0\hat{k})$

$\vec{a} \cdot (\vec{b} \times \vec{c}) = (3.0\hat{i} + 3.0\hat{j} - 2.0\hat{k}) \cdot (-8\hat{i} + 5\hat{j} + 6\hat{k})$

$= -24 + 15 - 12 = -21$

(b) $\vec{a} \cdot (\vec{b} + \vec{c}) = (3.0\hat{i} + 3.0\hat{j} - 2.0\hat{k}) \cdot (1.0\hat{i} - 2.0\hat{j} + 3.0\hat{k})$

$= 3 - 6 - 6 = -9$

(c) $\vec{a} \times (\vec{b} + \vec{c}) = (3.0\hat{i} + 3.0\hat{j} - 2.0\hat{k}) \times (1.0\hat{i} - 2.0\hat{j} + 3.0\hat{k})$

$= 5\hat{i} - 11\hat{j} - 9\hat{k}$